Failure-Contingent Timing and the Limits of a Stopping Component

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Abstract

We test a narrow component of physically timed robot recovery: deciding whether a retry can still finish within an original event-relative window. A success-only duration estimate overlooks the failed retry's observation wait and fallback. Accounting for this finite failure branch repairs one boundary case in 162 new MuJoCo control rollouts. A direct stopping baseline nevertheless achieves more timely outcomes. We retain that negative result, derive its explanation within the restricted dynamics, and specify the next test of conflicting necessary operations. This study establishes neither learned-policy gains nor chemical task success.

1. Question and conditional mechanism

A command acknowledgement does not determine when a physical process stops. After a failed command, a retry must be evaluated together with the cost of observing its failure and executing the fallback. For decision time t, reference event τ, retry duration d_r, fault-detection wait d_f, feedback transport d_o, relay delay d_g, and residual decay d_c, consider the two finite branches

$$D_{\text{success}} = d_r + d_c, \tag{1}$$

$$D_{\text{failure}} = d_r + d_f + d_o + d_g + d_c. \tag{2}$$

Their interval envelope $[D^-, D^+]$ gives the candidate completion support $[t^- + D^- - \tau^+,\ t^+ + D^+ - \tau^-]$. Containment in the required window is conditional on the action being executable and these duration bounds being valid. Our fixture assumes a reliable relay and only one retry. It does not provide a global recovery or safety guarantee. A positive decay lower bound measured at one initial speed is not transferred to arbitrary speeds; the candidate lower bound is zero.

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Table 1. Timing TRUE/FALSE/UNKNOWN over all 54 cases per control. Two earlier arms are retained comparators; the last three are new runs. The complete filter and scheduler deliberately share their decision rule.

Control	T	F	U
Ordinary retry (retained)	12	41	1
Single-action filter (retained)	17	36	1
Complete-branch filter	18	36	0
Matched branch scheduler	18	36	0
Direct relay	24	30	0

2. Controlled development experiment

We retain a contact-free, torque-driven rotor with a 10 ms native integration step. The reference is the physical shaft angle crossing 2 rad. Completion is the start of a segment with speed at most 0.1 rad/s persisting for 0.2 s. This is a flow surrogate, not a liquid or chemical endpoint. The 54 parameter cases cross three response settings, three feedback delays, two upper-deadline windows, and three command-fault conditions. Two settings are within the declared development calibration and one is a response shift outside it. These are not independent chemistry tasks or held-out confirmation trials.

We execute three new controls: complete-branch filtering, an explicitly matched deterministic scheduler using the same rule, and a direct relay at the first delivered trigger. All have identical action permissions and sensor access. The native model, transport, rendering and outcome definition remain unchanged; only candidate selection and the direct-relay baseline are added. The 162 fresh rollouts contain 129,762 state samples and 2,754 RGB frames. We reference 216 earlier controls, without counting them as newly executed data. Policies read delivered encoder and power packets, not the recorded RGB or hidden response/fault labels.

An independent persisted-data audit reconstructs every new outcome from raw samples, checks every frame digest, and verifies identical mechanical prefixes before the first accepted stop across the 54 comparator groups. It does not establish full-image pairing. Every planned parameter case, including failures and uncertain outcomes, stays in the denominator.

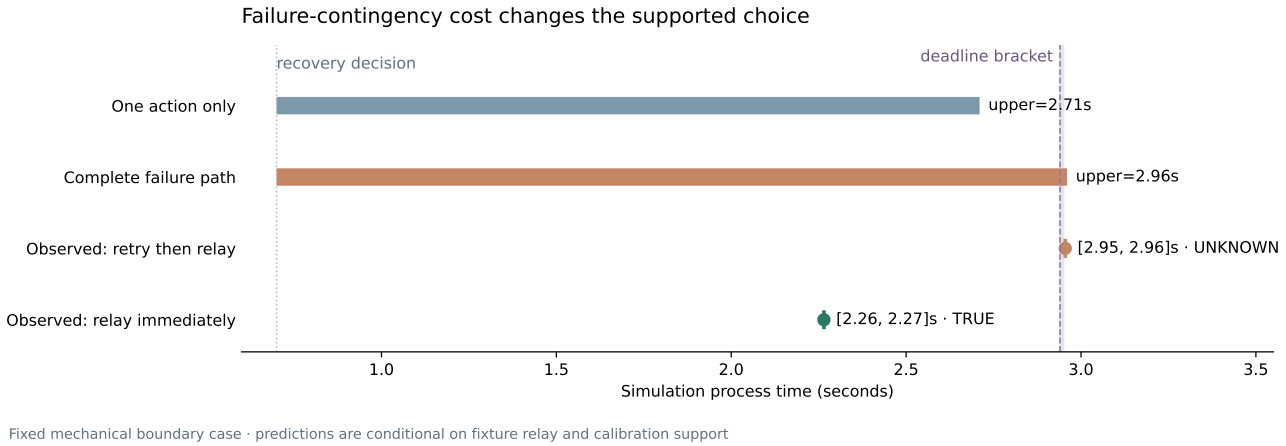


Figure 1. One repaired development case, in absolute simulation seconds. The original decision at 0.70 s predicts an upper completion time of 2.71 s only on retry success. The failed retry and fallback extend it to 2.96 s, beyond the deadline interval $[2.94, 2.95]$ s. The new control therefore selects the relay immediately. Actual endpoints are extracted independently and scored with the original reference and window.

3. Findings and the stronger baseline

The complete rule changes one uncertain boundary case to TRUE and changes no original TRUE case to a non-TRUE result. Its measured completion becomes $[2.26, 2.27]$ s, compared with $[2.95, 2.96]$ s under the retained single-action rule. Within the 36 calibration-range cases, complete filtering yields 17 TRUE, 19 FALSE and no UNKNOWN. Under response shift it yields only one TRUE in 18 cases. Four shifted parameter cases still produce an eligible prediction followed by a FALSE result; running both equivalent rules gives eight such trajectories. Finite calibration is not a coverage guarantee.

The direct relay yields 24 rather than 18 TRUE outcomes overall (21/36 in range and 3/18 under response shift). This comparison constrains our method claim. In this one-degree-of-freedom model, with inertia $I > 0$, damping $b > 0$, nonnegative drive u, and implicit velocity step Δt,

$$v_{k+1} = \frac{I v_k + \Delta t u_k}{I + \Delta t b}. \tag{3}$$

For equal initial states, pointwise smaller drive implies no larger velocity by induction. With a shared fixed reference, an upper deadline, and no other relay cost, earlier zero drive cannot delay the low-speed crossing. Thirty-six actual one-step probes match this formula to at most 1.78×10^{-15} absolute error; 18 native ordered-control pairs also respect the comparison. These are numerical mechanism checks, credited as zero additional benchmark rollouts. The argument does not apply to chemical endpoints, minimum-duration requirements, shared devices or nonlinear processes.

4. Evidence use and the next test

We export 162 audited offline event records with reference and completion intervals, simulation clock and raw-sample identities. A trusted external digest binds each full record; a digest alone is not source authentication. Online label-release time remains unmeasured. Consequently these post-hoc labels cannot enter online context, have zero robot behavior-cloning admissions, and assign zero chemical-return mask. A scoped offline mechanical value-target admission does not mean reinforcement learning was executed.

The next separately frozen development design introduces two processes on a shared bus: stopping A promptly with a global relay may prematurely terminate B's required sustained operation. Selective actions also compete for a serial command channel. We retain global stopping, ordinary selective retry, single-contract filtering and a matched joint-contract deterministic scheduler. This design has not been executed. Learning experiments must subsequently compare shared history-conditioned policy learning with equal-teacher-budget student-state labeling, and separate runtime restrictions, context updates and parameter updates. No VLA, diffusion-policy, RL or chemical-success gain is claimed by the present component study.